A Study of Controllable Pragmatics in Low-Resource Language Generation: Burmese with mT5

Group: AttentionIsAllYouNeed

Members

Thet Su Sann st126316

Thiri Shin Thant st126018

Hein Min Htet st126459

Tisa Bajracharya st126686

Motivation

Why Pragmatics Matters

Semantic ≠ Appropriate

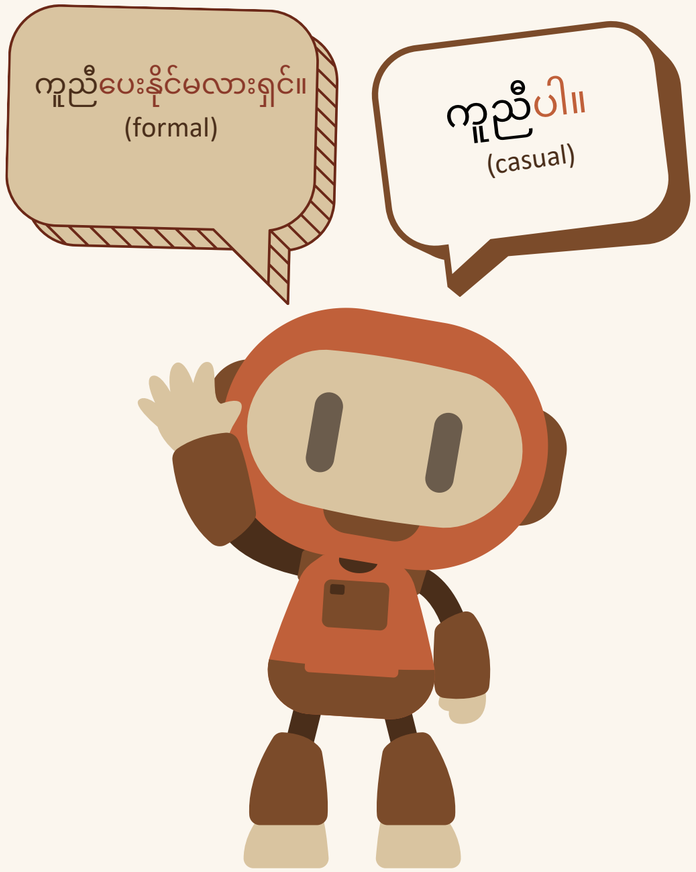
Semantic correctness alone does not guarantee appropriate language use in social contexts.

Pragmatics Defined

Pragmatics captures politeness, tone, and power dynamics — the social layer of language.

Burmese is Highly Sensitive

Burmese relies heavily on these pragmatic cues, making it a compelling case study.



Problem Statement

01

Focus on Meaning Only

Existing models prioritize semantic meaning but ignore sociolinguistic context entirely.

02

Low-Resource Languages Underexplored

Languages like Burmese receive little attention in research due to limited data.

03

No Fine-Grained Pragmatic Control

No current system supports controlling politeness, tone, and power in generation.

Research Goal

Study controllable generation
Analyze:

Context

Formal vs. casual settings

Politeness

High, medium, or low politeness

Tone

Direct, neutral, or indirect

Power Relations

Lower→higher, equal, higher→lower

Research Questions

RQ1	How does fine-tuning a multilingual text-to-text model affect its ability to generate Burmese utterances compared to a zero-shot baseline?
RQ2	To what extent does incorporating contextual information improve the quality and appropriateness of generated utterances?
RQ3	Which sociolinguistic attributes (e.g., politeness, tone, and power relations) contribute most to generation quality, and which are more challenging for the model to learn in a low-resource setting?

Dataset

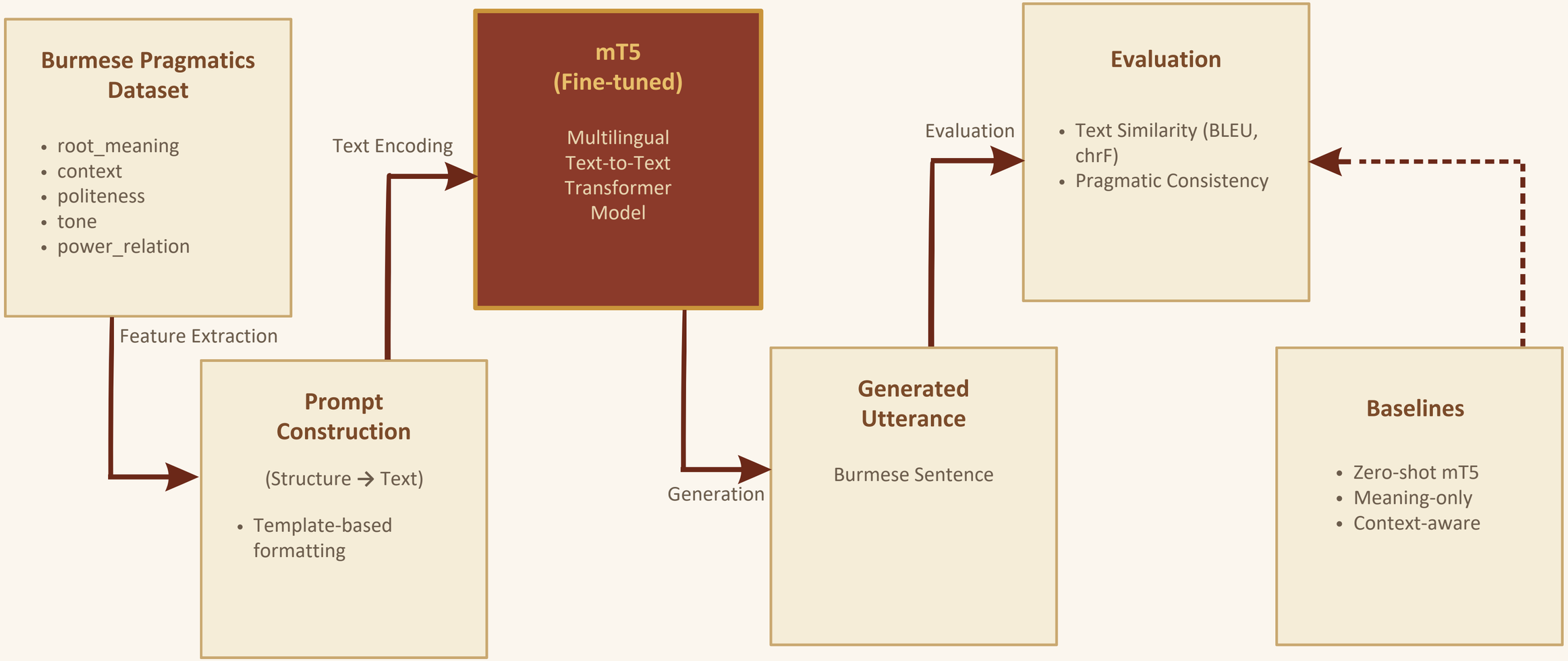
`freococo/burmese-contextual-pragmatics (HuggingFace)`

~2,200	22	4
Samples	Intents	Attributes

Attributes

- **context**
- **politeness**
- **tone**
- **power**

Method Overview



Format

Structured Pragmatic Input

Meaning: ask for help
Context: formal setting
Politeness: high
Tone: neutral
Power: lower → higher

Text Prompt

Generate Burmese Sentence:
Meaning: ask for help
Context: formal setting
Politeness: high
Tone: neutral
Power: lower → higher

mT5 (finetuned)

Generated Output (Burmese)

ကျေးဇူးပေးနိုင်မလားရှင်။

Properly polite and socially appropriate

Structured Pragmatic Input

Meaning: ask for help
Context: casual setting
Politeness: low
Tone: direct
Power: equal

Text Prompt

Generate Burmese Sentence:
Meaning: ask for help
Context: casual setting
Politeness: low
Tone: direct
Power: equal

mT5 (finetuned)

Generated Output (Burmese)

ကျေးဇူးပါ။

Direct / informal register

Experimental Design

Zero-shot mT5	Meaning-only	Context-aware	Full Model (Meaning+Pragmatics)

Input:
`"ask for help"`
	Input:
	`Meaning: ask for help`
		Input:
		`Meaning: ask for help`
		`Context: formal`
			Input:
			`Meaning: ask for help`
			`Context: formal`
			`Politeness: high`
			`Power: lower→higher`

Output: **ကူညီပါ။**	**Output:** **ကူညီပေးပါ။**	**Output:** **ကူညီပေးပါနော်။**	**Output:** **ကူညီပေးနိုင်မလားရှင်။**
(Direct / not polite enough)	*(Slightly improved, still generic)*	*(More polite, but incomplete)*	*(Properly polite and socially appropriate)*

*** These are illustrative examples. Final results will be validated through quantitative evaluation.***

Evaluation

01 Text Similarity

- BLEU
- chrF

02 Pragmatic Consistency

- Classifier
- Compare attributes

Analysis Plan

1

Attribute Ablation

Remove one attribute at a time to measure its individual impact on generation quality.

2

Attribute-Level Scores

Compute per-attribute metrics to identify which pragmatic features are easiest/hardest.

3

Error Analysis

Manually inspect failure cases to understand model weaknesses and patterns.

4

Rare Combinations

Examine performance on underrepresented attribute combinations in the dataset.

Expected Results

★

Full model performs best

Adding all pragmatic attributes (context, politeness, tone, power) will yield the highest generation quality.

≠

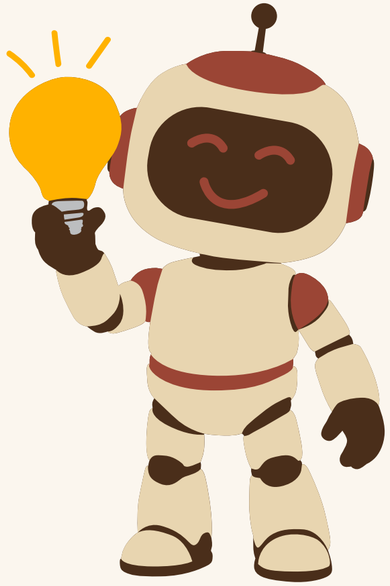
Not all attributes are equal

Some pragmatic dimensions will have stronger influence on output quality than others.

≉

BLEU ≠ Pragmatics

High lexical overlap scores may not correlate with sociolinguistically appropriate output - pragmatic consistency is a distinct dimension.



Limitations

Small Dataset

Only ~2,200 samples - insufficient for robust generalization across all pragmatic configurations.

Classifier Bias

The pragmatic consistency classifier may carry its own biases, affecting evaluation reliability.

Limited Generalization

Findings may be specific to the dataset distribution and may not transfer to broader Burmese language use.

Conclusion

- Controlled pragmatic generation
- Low-resource focus
- Analysis-driven study